

Tianruo Rose Xu

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Education

Cornell University, College of Engineering
B.S. in Computer Science Minor in Mathematics

Aug 2024 – May 2027 (expected)

Honors and Awards

- Hunter R. Rawlings III Cornell Presidential Research Scholar Award.
- Menschel Cornell Commitment Public Service Grant.
- Cornell Graduate School Opportunity Fund Grant.
- Clare Boothe Luce Undergraduate Research Award.
- Bowers Undergraduate Research Experience (BURE) stipend and Academic Year Expansion support.
- USA Computing Olympiad (USACO) Platinum Division; perfect score in the [January 2023 Gold Contest](#).
- USA Mathematical Olympiad (USAMO) Qualifier.

Publications

[Submitted] *RT-Safe: Benchmarking Safety of Agent in Real-Time Embodied Environment*.

[Submitted] *Can Emergent Verbalization of Epistemic Uncertainty Inform Factuality?*

[Preprint] Haoqiang Kang, Yiming Zhang, Yiyang Guo, Chuying Li, Jianzhi Shen, **Tianruo Rose Xu**, Xiaokang Ye, Lianhui Qin. *DeliveryGym: An RL Environment for Long-Horizon Embodied Agent Planning with Adaptive Curriculum*. arXiv preprint arXiv:2609.19801, September 2026. [\[Paper\]](#) [\[Code\]](#).

[Published] **Tianruo Rose Xu**. *Can You Trust What I Think? Analyzing and Improving Verbalized Uncertainty and Factuality in Reasoning-Based Large Language Models*. AAAI-26 Undergraduate Consortium, Singapore, January 2026. [\[Poster\]](#) [\[Oral Presentation\]](#) [\[Paper\]](#).

[Published] **Tianruo Rose Xu**, Vedant Gaur, Liu Leqi, Tanya Goyal. *The Progress Illusion: Revisiting Meta-Evaluation Standards of LLM Evaluators*. Findings of the Association for Computational Linguistics: EMNLP 2025, Suzhou, China, November 2025. [\[Paper\]](#) [\[Poster\]](#).

[Published] Yingke Ding, Jiankai Tang, Wanying Mo, **Tianruo Rose Xu**, Yuanchun Shi, Yuntao Wang. *NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model*. MICCAI 2025 Workshop MSB EMERGE, July 21, 2025. [\[Paper\]](#) [\[Code\]](#). Runner-up Best Paper, MICCAI 2025 Workshop MSB EMERGE.

[Published] Tianruo Xu. *Elder Depression Detection by Multimodal Means*. BCP Education & Psychology, November 7, 2022. [\[Paper\]](#).

Research Experience

Undergraduate Researcher, Cornell University

Computer Science Department, NLP Group; advised by Professor Tanya Goyal and Claire Cardie.

Analyzing Verbalized Uncertainty and Factuality in Reasoning-Based LLMs 2025 – 2026

- Investigates whether verbalized epistemic uncertainty in LLM reasoning traces predicts fact-level factuality in long-form question answering.
- Led project design with Anirudh Atmakuru and Sanjana Ramprasad; built a pipeline for trace analysis, atomic fact extraction, trace-response alignment, FactScore evaluation, large-scale empirical analysis, and classifier training.

Revisiting Meta-Evaluation Standards of LLM Evaluators 2025

- Develops fine-grained meta-evaluation methods for auditing LLM-as-judge benchmarks, evaluator reliability, and reference-model sensitivity.
- Led the evaluation framework design; implemented data collection and scoring pipelines; ran large-scale experiments; analyzed correlations with human judgments; contributed to writing, figures, and result interpretation.

Content and Style Evaluation in Natural Language Generation Systems 2024

- Studies content and style as separable evaluation dimensions in natural language generation systems.

- Co-supervised by Professor Claire Cardie; designed experiments, implemented data processing and evaluation pipelines, ran model-based evaluations, conducted error analysis and ablations, and contributed to writing and result visualization.

Research Intern, UC San Diego

Computer Science Department, Q-Lab; advised by Professor Lianhui Qin.

RT-Safe: Real-Time Embodied Agent Safety 2026

- Lead RT-Safe, a benchmark evaluating embodied agent safety in dynamic environments that continue evolving during model inference; manuscript submitted.
- Defined the research direction, safety scenario taxonomy, benchmark tasks, and evaluation workflow; led analysis of how inference latency and reasoning effort affect navigation safety.

DeliveryGym / DeliveryBench: Embodied Agent Environments 2026

- Developed visual traffic, obstacle, and safety infrastructure for VAGEN/DeliveryBench, including pedestrian traffic-light rendering, signal-state checks, rollout support, and hazard evaluation tools.
- Built and maintained the Unreal Engine-rendered first-person-view observation pipeline and assets, including traffic-signal and blocked-road variants.
- Extended DeliveryBench with the CityCore Paris map, integrating semantic points of interest, buildings, road and navigation graphs, coordinate transforms, and 2D map assets; contributed CityCore Paris visual materials.

Undergraduate Researcher, Cornell University

Mathematics Department, Math + AI Lab; advised by Professor Daniel Halpern-Leistner.

Lean Proof Arena and LaTeX-to-Lean Autoformalization Pipeline 2026

- Develops a DARPA ExPMath-supported platform and pipeline for research-level autoformalization, centered on Lean Proof Arena and LaTeX-to-Lean conversion.
- Building Lean Proof Arena for comparing formal proof attempts; developing a LaTeX-to-Lean pipeline, benchmark tasks, and evaluation protocols for formal correctness and semantic fidelity.

Undergraduate Research Assistant, Tsinghua University

Department of Computer Science and Technology, Pervasive Human-Computer Interaction Lab.

Intelligent AI Tool Benchmarking and Orchestration Platform 2025 – 2026

- Develops an intelligent AI tool platform that standardizes tool capability descriptions, matches tools to user intents, and supports multi-tool workflow orchestration.
- Designed and implemented back-end search and matching algorithms for tool retrieval, user-intent matching, and personalized AI tool selection.

NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model 2025

- Develops a multi-task learning model for automated nailfold capillaroscopy, combining segmentation, classification, and keypoint detection for non-invasive microvascular image analysis.
- Conducted dataset preprocessing, baseline training, experiment execution, model evaluation, and preliminary result analysis to support development of the NFCMTL framework.

Undergraduate Research Assistant, Cornell University 2024

Information Science Department; advised by Keigo Kusumegi and Professor Yian Yin.

- Studied how diverse collaboration patterns affect Wikipedia content quality.
- Processed Wikipedia data and applied K-means clustering to analyze article-quality patterns.

Teaching Experience

Cornell University

- **Head Teaching Assistant**, CS 3780: Introduction to Machine Learning Fall 2026
- **Climbing Instructor**, PE 1640: Basic Rock Climbing Fall 2026
- **Teaching Assistant**, CS 3780: Introduction to Machine Learning Spring 2026
Designed the final Kaggle challenge, [All About Birds!](#), a bird-sound spectrogram classification task.
- **Teaching Assistant**, CS 4820: Introduction to Analysis of Algorithms Fall 2025
- **Teaching Assistant**, CS 2800: Mathematical Foundations of Computing Spring 2025